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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

16 October 2026

Hiring Team
Faculty

Application: Senior Machine Learning Engineer (Life Sciences)

Dear Faculty hiring team,

I am applying for the Senior Machine Learning Engineer role in Faculty's Life Sciences team. For three years at HDFC Asset Management I owned machine learning and GenAI systems end to end - scoping the architecture, building the software, deploying it for real users and fixing it in production - and doing that same work on therapeutic and scientific data is what I want to do next. The Life Sciences team's mix of deep Python engineering, architectural ownership and client advisory is the shape of the work I have been doing, and I would like to bring it to Faculty.

At HDFC I designed and built the applications end to end: Python FastAPI microservices and a ReactJS frontend, retrieval with LlamaIndex and Azure OpenAI GPT-3.5 and Llama-3, a separate pgvector database per department, and agentic tool-calling where agents call Python functions to run SQL-backed comparisons and generate PDF reports on request. I packaged the services as Docker containers and shipped them on AWS (EC2 including a GPU instance, S3, ECR and Lambda) through a one-click Jenkins and GitLab CI/CD pipeline, taking the applications from repeatedly failing the security team's VAPT to passing it. Where the model choice was an engineering trade-off rather than a default, I self-hosted Llama-3 on a GPU EC2 instance for our largest department to control cost and kept the stronger Azure OpenAI outputs for the decision makers.

The closest work I have done to Faculty's mission is my MSc dissertation with AstraZeneca: CellLineSelector, a tool that ranks the best-suited cancer cell lines for a target by inclusion and exclusion genes. I designed the scoring and ranking algorithm, the preprocessing pipeline over DepMap, GEO and HPA data and the RAG-based assistant that narrates results and answers methodology questions, and we refined and validated the ranking with feedback from AstraZeneca scientists. It was a team project that I co-built, and it is the project that convinced me I want to keep building production AI for life sciences.

I want to be direct about where my experience differs from your list. My production work is Python services and GenAI rather than models built in TensorFlow or PyTorch: my deep learning is academic, the MSc multi-architecture study and a CNN project in my B.Tech, and I have not trained or served models of that kind in production. I have not worked with Kubernetes (Docker only), and my monitoring is production troubleshooting and user feedback rather than drift detection, evaluation harnesses or model observability tooling; at HDFC quality was held by system-prompt guardrails and Compliance review. My technical advisory has been internal, to Client Service, Compliance and Investment stakeholders, not to external pharma or MedTech clients, and my experience is about three years rather than the significant track record the posting describes, though those years were spent owning systems end to end.

I am based in Bristol and happy to discuss the hybrid arrangement in London. I hold a Student Visa and am eligible for the Graduate Visa, which gives two years of full working rights. Thank you for considering my application; I would be glad to talk it through.

Yours sincerely,
Kaustubh Sonawane